

Exercise 1. Use proof by reduction to show that in the AGM, if the DL assumption holds for a group, then so does the CDH assumption. Recall that an adversary that succeeds in the CDH attack game in the AGM also provides a list of exponents!

Let G be a cyclic group of order q with generator g . Assume the DL assumption holds for G .

Suppose, for contradiction, that there exist a PPT algebraic adversary $\mathcal{A}_{\text{CDH}}$ which solves AGM-CDH problem with non-negligible probability ϵ .

We solve this by constructing a reduction that solves the DL problem with non-negligible probability using $\mathcal{A}_{\text{CDH}}$ as subroutine.

Let (g, X) with $X = g^x$ be a given DL challenge.

1. Select $t \in \mathbb{Z}_q$ uniformly at random and set $Y = g^t$.
2. Run $\mathcal{A}_{\text{CDH}}$ on input (g, X, Y) to obtain (Z, z) .

$$z = (a, b, c) \in \mathbb{Z}_q^3 \text{ and } Z = g^a X^b Y^c.$$

3. If $Z \neq g^{xt}$, abort.

4. Otherwise, since $X = g^x$ and $Y = g^t$, we have

$$Z = g^a X^b Y^c = g^a (g^x)^b (g^t)^c = g^{a+bx+ct} = g^{xt}$$

$$\text{So } a+bx+ct = xt \pmod{q}$$

$$\Rightarrow a+ct = x(t-b)$$

If $b \neq t$, we can solve for x ,

$$x = (a+ct)(t-b)^{-1} \pmod{q}$$

Otherwise abort

5. Output x .

The reduction runs $\mathcal{A}_{\text{CDH}}$ once.

If $b=t$, the probability that the reduction aborts is exactly $1/q$, since t is random and independent of b .

$$\therefore \Pr[\text{DL}^{\mathcal{A}}(n)=1] \geq \Pr[\text{CDH}^{\mathcal{A}}(n)=1] - 1/q = \epsilon - 1/q$$

ie, if $\mathcal{A}_{\text{CDH}}$ solves CDH with probability ϵ , the reduction solves DL with probability at least $\epsilon - 1/q$

$\Rightarrow$ If CDH is easy in AGM, then so is DL, contradicting the DL assumption. Thus CDH is also hard in the AGM.

Exercise 2. Use proof by reduction in the AGM to show that if the DL assumption holds, then the Schnorr signature is EUF-CMA secure.

PROOF BY REDUCTION

Let F be a PPT algebraic adversary that, after access to the signing oracle and random oracle, outputs a forgery (m^*, e^*, s^*) with non-negligible probability.

Let $(g, y = g^x)$ be a DL instance.

Construct an algorithm A as follows:

- Simulate the Schnorr system for F using only g, y
- Whenever a signature query for a message m is requested:
 - Pick sample random k , compute $u = g^k$
 - Program $H(u, m) = e$ for random e .
 - Respond with $s = k + ex \pmod q$
- For random oracle queries, ensure answers are consistent with previous answers and signatures.

Eventually, F outputs a forgery: message m^* , signature (e^*, s^*) where $e^* = H(u^*, m^*)$ and $g^{s^*} = u^* y^{e^*}$

In the AGM, F must also output an explicit representation.

$$u^* = g^\alpha y^\beta \quad (\alpha, \beta \in \mathbb{Z}_q)$$

Substitute the algebraic representation for u^* into the verification equation:

$$\begin{aligned} g^{s^*} &= (g^\alpha y^\beta) y^{e^*} = g^\alpha \cdot y^{\beta + e^*} = g^\alpha \cdot (g^x)^{\beta + e^*} \\ &= g^{\alpha + x(\beta + e^*)} \end{aligned}$$

Hence,

$$g^{s^*} = g^{\alpha + x(\beta + e^*)} \Rightarrow s^* = \alpha + x(\beta + e^*) \pmod q$$

$$\text{So, } s^* - \alpha = x(\beta + e^*) \pmod{q}$$

If $\beta + e^* \not\equiv 0 \pmod{q}$, then

$$x \equiv (s^* - \alpha)(\beta + e^*)^{-1} \pmod{q}$$

Thus $\mathcal{A}$ has solved discrete log for x .

The probability that $\beta + e^* \equiv 0 \pmod{q}$ is $1/q$ and is negligible since β is uniform in $\mathbb{Z}_q$ independent of the random e^* . Thus $P[\beta] = 1/q$ (negligible).

Otherwise $\mathcal{A}$ outputs the correct DL with probability $\geq \epsilon - 1/q$.

$$\text{Hence } P_u[DL^{\mathcal{A}}(n) = 1] \geq P_e[\text{forge}^F = 1] - 1/q = \epsilon - 1/q$$

If DL is hard, then ϵ must be negligible.

i.e., F cannot forge with non-negligible probability.

$\therefore$ If there is a efficient forger in AGM-ROM, there is an efficient DL solver, conflicting with the DL hardness assumption.

Thus, in the AGM, the security of Schnorr signatures (against algebraic, EUF-CMA forgers in the AGM-ROM) reduces tightly to the hardness of the DL problem.

The tightness and simplicity of the reduction are direct consequences of the algebraic requirements in AGM.

$\therefore$ Assuming DL is hard, Schnorr signatures are EUF-CMA secure in AGM-ROM.

Exercise 2. Use proof by reduction in the AGM to show that if the DL assumption holds, then the Schnorr signature is EUF-CMA secure.

PROOF BY REDUCTION

Let F be a PPT algebraic adversary that, after access to the signing oracle and random oracle, outputs a forgery (m^*, R_i^*, s^*) with non-negligible probability.

Let $(g, y = g^x)$ be a DL instance.

Construct an algorithm A as follows:

- Simulate the Schnorr system for F using only g, y
- Whenever a signature query for a message m is requested:
 - Pick sample random $k_i \xleftarrow{\$} \mathbb{Z}_q$, set $R_i = g^{k_i}$
 - Program the random oracle so that $H(R_i, m_i) = e_i$ for random e_i
 - Return signature (e_i, s_i) with $s_i = k_i + e_i x \pmod{q}$
 - Store pairs (R_i, k_i) for use in extraction.
- For random oracle queries, ensure answers are consistent with previous answers and signatures.

For hash queries on pairs $(\tilde{R}_j, m_j)$:

- Each $\tilde{R}_j$ must come with an algebraic representation.
$$\tilde{R}_j = g^{a_0} y^{a_1} \prod_{i=1}^q R_i^{a_{i+1}}$$
- If $(\tilde{R}_j, m_j)$ matches a prior programmed pair, respond consistently.
- Otherwise, for new $(\tilde{R}_j, m_j)$, reply with $e_j \xleftarrow{\$} \mathbb{Z}_q$, recording this assignment.

Eventually, F outputs a forgery: (m^*, R^*, s^*)

$e^* = H(R^*, m^*)$ and $g^{s^*} = R^* y^{e^*}$ to verify

By AGM, R^* is accompanied by exponents:

$$R^* = g^{a_0} y^{a_1} \prod_{i=1}^q R_i^{a_{i+1}}$$

Substituting R^* and k_i , we get

$$\begin{aligned} g^{s^*} &= R^* y^{e^*} = g^{a_0} y^{a_1} \prod_{i=1}^q R_i^{a_{i+1}} y^{e^*} \\ &= g^{a_0} g^{x a_1} \prod_{i=1}^q g^{k_i a_{i+1}} g^{x e^*} \\ &= g^{a_0 + x a_1 + \sum_{i=1}^q (k_i a_{i+1} + x e^*)} \\ &= g^{a_0 + \sum_{i=1}^q k_i a_{i+1} + x(a_1 + e^*)} \end{aligned}$$

$$\therefore s^* \equiv a_0 + \sum_{i=1}^q k_i a_{i+1} + x(a_1 + e^*) \pmod{q}$$

$$\Rightarrow s^* - a_0 - \sum_{i=1}^q k_i a_{i+1} \equiv x(a_1 + e^*) \pmod{q}$$

$$\Rightarrow x \equiv (s^* - a_0 - \sum_{i=1}^q k_i a_{i+1}) \cdot (a_1 + e^*)^{-1} \pmod{q}$$

as long as $a_1 + e^* \not\equiv 0 \pmod{q}$

Output x as the solution to the DH instance.

Thus $\mathcal{A}$ has solved discrete log for x .

The probability that $a_1 + e^* \equiv 0 \pmod{q}$ is $1/q$ and is negligible since a_1 is uniform in $\mathbb{Z}_q$ independent of the random e^* . Thus $P[a_1] = 1/q$ (negligible).

Otherwise A outputs the correct DL with probability $\geq \epsilon - 1/q$

$$\text{Hence } P_A[DL^A(n)=1] \geq P_F[\text{forge}^F=1] - 1/q = \epsilon - 1/q$$

If DL is hard, then ϵ must be negligible.

i.e., F cannot forge with non-negligible probability.

$\therefore$ If there is a efficient forger in AGM+ROM, there is an efficient DL solver, conflicting with the DL hardness assumption.

Thus, in the AGM, the security of Schnorr signatures (against algebraic, EUF-CMA forgers in the AGM-ROM) reduces tightly to the hardness of the DL problem.

The tightness and simplicity of the reduction are direct consequences of the algebraic requirements in AGM

$\therefore$ Assuming DL is hard, Schnorr signatures are EUFCMA secure in AGM-ROM.

Exercise 2. Use proof by reduction in the AGM to show that if the DL assumption holds, then the Schnorr signature is EUF-CMA secure.

Let $\mathcal{A}'$ be a probabilistic polynomial time adversary against the Schnorr signature scheme in the AGM-ROM setting.

The adversary $\mathcal{A}'$ receives the public key $pk = y = g^x$ for unknown secret $x \in \mathbb{Z}_q$, has access to signing oracle and a random oracle H .

The goal is to construct an algorithm $\mathcal{A}$ that solves the discrete logarithm problem in G using $\mathcal{A}'$ as a subroutine.

Algorithm $\mathcal{A}$ is given as input the DH challenge $(g, y = g^x)$ and simulate the Schnorr EUF-CMA experiment for $\mathcal{A}'$ as follows.

First $\mathcal{A}$ maintains a list of previously answered signing queries and their associated signatures as well as the programmed hash oracle responses.

To simulate a signing query on message m_i , $\mathcal{A}$ picks two random elements $s_i, e_i \in \mathbb{Z}_q$ uniformly at random.

Then it sets,

$$R_i := g^{s_i} y^{-e_i}$$

By construction, $g^{s_i} = R_i y^{e_i}$ so the signature (e_i, s_i) passes verification with respect to the message m_i .

Algorithm $\mathcal{A}$ programs the hash oracle H such that $H(R_i, m_i) = e_i$ and responds to the signing query with (e_i, s_i) .

When the adversary $\mathcal{A}'$ queries the random oracle H on inputs $(\tilde{R}_j, m_j)$, algorithm $\mathcal{A}$ answers consistently: if $(\tilde{R}_j, m_j)$ matches a previously programmed pair, $\mathcal{A}$ returns the same e_j , otherwise it returns a fresh uniform random e_j and records the query.

The adversary $\mathcal{A}'$ must output a forgery (m^*, e^*, s^*) for some messages m^* not previously queried to the signing oracle.

The forgery must satisfy $g^{s^*} = R^* y^{e^*}$
 where $e^* = H(R^*, m^*)$

Because we are in the AGM, the adversary must also provide an algebraic representation of the group element R^* as a linear combination of known elements.

$$R^* = g^{a_0} y^{a_1} \prod_{i=1}^q R_i^{a_{i+1}}$$

for some known integer exponents $a_0, a_1, \dots, a_{i+1} \in \mathbb{Z}_q$

$\therefore$ Verification equation becomes

$$g^{s^*} = R^* y^{e^*} = g^{a_0} y^{a_1} \prod_{i=1}^q R_i^{a_{i+1}} y^{e^*}$$

Since $R_i = g^{s_i} y^{-e_i}$, for each i , we substitute to obtain

$$g^{s^*} = g^{a_0} y^{a_1} \prod_{i=1}^q (g^{s_i} y^{-e_i})^{a_{i+1}} y^{e^*}$$

$$= g^{a_0} y^{a_1} \prod_{i=1}^q g^{s_i a_{i+1}} y^{-e_i a_{i+1}} y^{e^*}$$

$$= g^{a_0} y^{a_1} g^{\sum_{i=1}^q s_i a_{i+1}} y^{-\sum_{i=1}^q e_i a_{i+1}} y^{e^*}$$

$$= g^{a_0 + \sum_{i=1}^q s_i a_{i+1}} y^{a_1 + e^* - \sum_{i=1}^q e_i a_{i+1}}$$

Since $y = g^x$, we have

$$\begin{aligned} g^{s^*} &= g^{a_0 + \sum s_i a_{i+1}} \cdot g^{x(a_1 + e^* - \sum e_i a_{i+1})} \\ &= g^{a_0 + \sum s_i a_{i+1} + x(a_1 + e^* - \sum e_i a_{i+1})} \end{aligned}$$

Equating exponents modulo q , we have

$$s^* \equiv a_0 + \sum_{i=1}^q s_i a_{i+1} + x \left(a_1 + e^* - \sum_{i=1}^q e_i a_{i+1} \right) \pmod{q}$$

$$\Rightarrow x \equiv \left(s^* - a_0 - \sum_{i=1}^q s_i a_{i+1} \right) \left(a_1 + e^* - \sum_{i=1}^q e_i a_{i+1} \right)^{-1} \pmod{q}$$

Provided the denominator is nonzero modulo q .

Since the values e_i, e^* and the adversary's coefficient a_i are essentially random, the probability that the denominator is zero mod q is negligible (i.e., on the order of $1/q$).

Thus $\mathcal{A}$ extracts the discrete logarithm x .

Since the discrete logarithm problem is assumed hard, no such adversary $\mathcal{A}$ can exist with non-negligible forging probability.

This completes the proof of Schnorr's EUF-CMA security in the AGM+ROM.

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Exercise 3. Many applications of Sigma protocols require a large challenge space. This exercise shows that we can always take a Sigma protocol with a small challenge space and turn it into one with a large challenge space, essentially by parallel repetition.

Let (P, V) be a Sigma protocol for a relation $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{Y}$, with challenge space $\mathcal{C}$. Let k be a positive integer. Define a new Sigma protocol (P^k, V^k) as follows. Here, the prover P^k takes as input $(x, y) \in \mathcal{R}$, the verifier V^k takes as input $y \in \mathcal{Y}$, and the challenge space is $\mathcal{C}^k$.

- P^k initializes k instances of P on input (x, y) , obtaining commitments $t_1, \dots, t_k$, and sends these to V^k .
- V^k chooses $(c_1, \dots, c_k) \in \mathcal{C}^k$ at random, and sends this to P^k .
- For $i = 1, \dots, k$ the prover P^k feeds c_i into the i th instance of P , obtaining a response z_i . It sends $(z_1, \dots, z_k)$ to V^k .
- For $i = 1, \dots, k$ the verifier V^k verifies that (t_i, c_i, z_i) is an accepting conversation for y .

(a) Show that (P^k, V^k) is a Sigma protocol for $\mathcal{R}$.

To show that (P^k, V^k) satisfies the conditions of a Sigma protocol, we verify the following.

1. PROTOCOL STRUCTURE:

The interaction between P^k and V^k has three moves characteristic of Sigma protocol:

- Commit: The tuple of commitments $(t_1, t_2, \dots, t_k)$ is sent by P^k to V^k .
- Challenge: V^k responds with the challenge vector $(c_1, c_2, \dots, c_k)$ sampled uniformly from $\mathcal{C}^k$.
- Response: P^k replies with the vector of responses $(z_1, z_2, \dots, z_k)$.

Verification depends deterministically on y and

$(t_1, \dots, t_k), (c_1, \dots, c_k), (z_1, \dots, z_k)$ since each (t_i, c_i, z_i) is verified as in (P, V) .

2. COMPLETENESS: Since (P, V) is a Sigma Protocol for R , it satisfies perfect completeness.

$\forall (x, y) \in R$, the honest prover $P(x, y)$ and verifier $V(y)$ always produce accepting conversations (t, c, z) .

Running k independent copies means each of the k parallel sessions individually produces an accepting transcript with probability 1

- V^k chooses $(c_1, \dots, c_k) \in \mathcal{C}^k$ at random, and sends this to P^k .
- For $i = 1, \dots, k$ the prover P^k feeds c_i into the i th instance of P , obtaining a response z_i . It sends $(z_1, \dots, z_k)$ to V^k .
- For $i = 1, \dots, k$ the verifier V^k verifies that (t_i, c_i, z_i) is an accepting conversation for y .

(a) Show that (P^k, V^k) is a Sigma protocol for $\mathcal{R}$.

(b) Show that if (P, V) provides knowledge soundness, then so does (P^k, V^k) .

(c) Show that if (P, V) is special HVZK, then so is (P^k, V^k) .